Empirical Evaluation of On-Device Small Language Models on Consumer Hardware for Privacy-Compliant University Workflows

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Abstract

Higher education institutions face increasing pressure to automate administrative workflows while strictly adhering to statutory privacy mandates under the Family Educational Rights and Privacy Act (FERPA) and Title II of the Americans with Disabilities Act (ADA). While proprietary cloud-hosted Large Language Models (LLMs) offer high generative fluencies, their deployment introduces critical data-perimeter risks and recurring subscription overheads. On-device Small Language Models (SLMs) running locally on commodity enterprise workstations present a viable alternative; however, their capacity to execute rigorous institutional text workflows remains under-characterized across extended parameter variations. In this study, we empirically evaluate two open-weight models (`llama3.2:3b` and `qwen2.5:7b`) deployed on a commodity enterprise workstation (Intel Core i7, 16 GB RAM, CPU-only execution) across an expanded parameter matrix: quantization precision (`Q4_K_M` vs. `Q8_0`), decoding temperature ablations ($\in [0.0, 0.2, 0.7]$), token latency distributions (p50/p95/p99), automated ROUGE/BERTScore policy metrics, and mathematical Human Intervention Rate (HIR) labor modeling. Our findings demonstrate that while 3B models achieve higher generation throughput (108 tokens/sec), they fail strict JSON schema primitive type casting and exhibit high hallucination rates. Conversely, 7B models achieve 100% schema validity, robust lexical coverage (ROUGE-L > 0.82), and sustainable administrative overhead.

1 Introduction

Higher education institutions face conflicting pressures at the intersection of operational scaling and statutory compliance. Expanding administrative workloads—including student helpdesk triage, syllabus compliance auditing, and policy communication—present natural use-cases for generative artificial intelligence. However, standard commercial implementations relying on proprietary cloud APIs introduce substantial data governance vulnerabilities under FERPA and institutional data-perimeter mandates [1]. Transmitting student inquiries, grade dispute appeals, and unvetted course materials beyond campus boundaries creates unacceptable institutional exposure.

Recent advancements in quantization and small language models (SLMs) spanning 3B to 7B parameters suggest that edge deployment on commodity x86 office laptops is theoretically feasible [2, 3]. Nevertheless, significant empirical questions remain regarding how quantization depth, decoding temperatures, and latency distribution percentiles impact model stability in constrained settings.

This paper extends prior edge evaluation frameworks by incorporating five rigorous experimental dimensions:

(1) **Quantization Precision Ablation** comparing 4-bit medium (`Q4_K_M`) against 8-bit (`Q8_0`) memory footprints.
(2) **Decoding Parameter Sensitivity** analyzing temperature variance ($\in [0.0, 0.2, 0.7]$) on structural constraint retention.
(3) **Token-Level Latency Distribution Profiling** capturing p50, p95, and p99 latency percentiles to assess CPU thermal throttling.
(4) **Automated Lexical and Semantic Evaluation** computing ROUGE-L and BERTScore for policy summarization fidelity.
(5) **Mathematical Human Intervention Rate (HIR) Cost Modeling** quantifying administrative labor savings versus verification overhead.

2 Related Work

2.1 Edge & Consumer Hardware Model Inference

Benchmarking local language models on resource-constrained hardware has gained significant momentum as edge quantization formats mature. Husom et al. [2] systematically evaluated quantized open models in Ollama, establishing that consumer-grade processors can sustain viable operational latencies when properly configured. Similarly, Kurt [3] demonstrated that 4-bit Medium (`Q4_K_M`) quantization maintains approximately 99

2.2 Institutional Privacy & FERPA Compliance

Cloud-based AI architectures present unresolved compliance hurdles in higher education. Dong Liebowitz [1] identified that commercial LLM APIs inherently expand an institution's data perimeter, complicating compliance with FERPA data-privacy protections. Operating models on-premises guarantees that course evaluations, student grievances, and accommodation requests never exit institutional control.

3 Methodology & Extended Parameter Framework

3.1 Hardware Environment & Deployment Stack

Experiments were conducted on an enterprise-standard x86 mobile workstation: Intel Core i7 CPU (16 cores, base AVX2 instruction set), 16 GB DDR5 RAM, and integrated graphics (CPU-only execution). Local inference was orchestrated via Ollama serving quantized weights using default `llama.cpp` CPU thread allocations.

3.2 Extended Evaluation Parameters

To rigorously characterize operational viability, our evaluation harness records:

- **Memory Quantization Scaling:** Evaluating base RAM allocation across `Q4_K_M` and `Q8_0` formats.
- **Decoding Ablations:** Testing temperature settings $\in \{0.0, 0.2, 0.7\}$ to measure output determinism and syntax collapse.
- **Latency Percentiles:** Capturing p50, p95, and p99 generation latencies to identify memory bus saturation.
- **Automated Quality Metrics:** Measuring ROUGE-1, ROUGE-2, ROUGE-L, and BERTScore against gold-standard references.

4 Empirical Results Advanced Parameter Analysis

4.1 Quantization Depth and Latency Percentiles

Table 1 summarizes performance across quantization levels and latency distributions.

4.2 Detailed Error Taxonomy Failure Modes

Qualitative inspection reveals distinct failure modalities between sub-4B and 7B tiers:

- **Primitive Type Collapse:** Sub-4B models frequently output grading breakdown weights as strings (e.g., `"30%"`) instead of integer or float primitives, causing downstream JSON schema validation failures.
- **Conversational Preamble Leakage:** When subjected to zero-shot formatting constraints, smaller models emit conversational wrappers (e.g., *'Here is your requested JSON:"*), violating strict parsing pipelines.

4.3 Automated Policy Summarization (ROUGE BERTScore)

Evaluating policy summaries against reference ground truth yields strong semantic alignment for 7B models:

- `llama3.2:3b`: ROUGE-L = .741, $BERTScore() = .862$.
- `qwen2.5:7b`: ROUGE-L = **0.835**, BERTScore () = **0.914**.

4.4 Human Intervention Rate (HIR) Cost Modeling

To quantify institutional labor impact, we model administrative net savings as:

$$\text{Net Savings} = (\text{Task Volume} \times t_{\text{manual}}) - (\text{Task Volume} \times \text{HIR} \times t_{\text{correction}}) \quad (1)$$

Where $\text{HIR}_{3b} = 1.0$ (complete re-authoring required due to hallucinations) and $\text{HIR}_{7b} = 0.15$ (minor slot-filling verification).

5 Institutional Governance & Policy Implications

Integrating 7B open-weight models under rigorous data-perimeter constraints ensures full FERPA and ADA Title II compliance while maintaining positive net administrative labor savings. Furthermore, local execution mitigates third-party data-use agreement friction.

6 Conclusion

This study expanded local LLM benchmarking across quantization depth, latency distributions, and automated quality metrics, establishing that 7B models at `Q4_K_M` represent the optimal deployment tier for privacy-compliant higher education workflows.

Author Contributions (CRediT Statement)

Mohammed Iliyas Sanjamala: Conceptualization, Methodology, Software, Investigation, Formal Analysis, Writing – Original Draft, Project Administration. **Karthikeyan Prem Kumar:** Data Curation, Validation, Investigation, Writing – Review & Editing. **Richard Smith:** Conceptualization (Higher Education Context), Validation, Supervision (Institutional Domain), Writing – Review & Editing. **Aditi Singh:** Supervision (Technical Rigor), Methodology, Formal Analysis, Writing – Review & Editing.

References

[1] Y. Dong and J. Liebowitz. 2025. Minimizing Data Exposure in Higher Education LLM Applications. *Brandeis University Technical Working Paper* (2025).
[2] Erik Husom et al. 2025. Sustainable LLM Inference for Edge AI: Evaluating Quantized LLMs for Energy Efficiency, Output Accuracy, and Inference Latency. *arXiv preprint arXiv:2504.03360* (2025).
[3] B. Kurt. 2026. Which Quantization Should I Use? A Unified Evaluation of llama.cpp Quantization Formats. *arXiv preprint arXiv:2601.14277* (2026).

Table 1: Extended Empirical Matrix: Quantization, Throughput, and Latency Percentiles

Model	Quant	RAM	Throughput	TTFT	p50 Latency	p95 Latency	p99 Latency
`llama3.2:3b`	`Q4_K_M`	2.2 GB	108.4 tps	0.04s	9.2 ms	14.5 ms	22.1 ms
`llama3.2:3b`	`Q8_0`	3.8 GB	84.1 tps	0.06s	11.8 ms	19.3 ms	28.4 ms
`qwen2.5:7b`	`Q4_K_M`	4.9 GB	54.2 tps	0.08s	18.4 ms	29.1 ms	42.6 ms
`qwen2.5:7b`	`Q8_0`	8.2 GB	38.6 tps	0.14s	26.2 ms	41.5 ms	59.8 ms